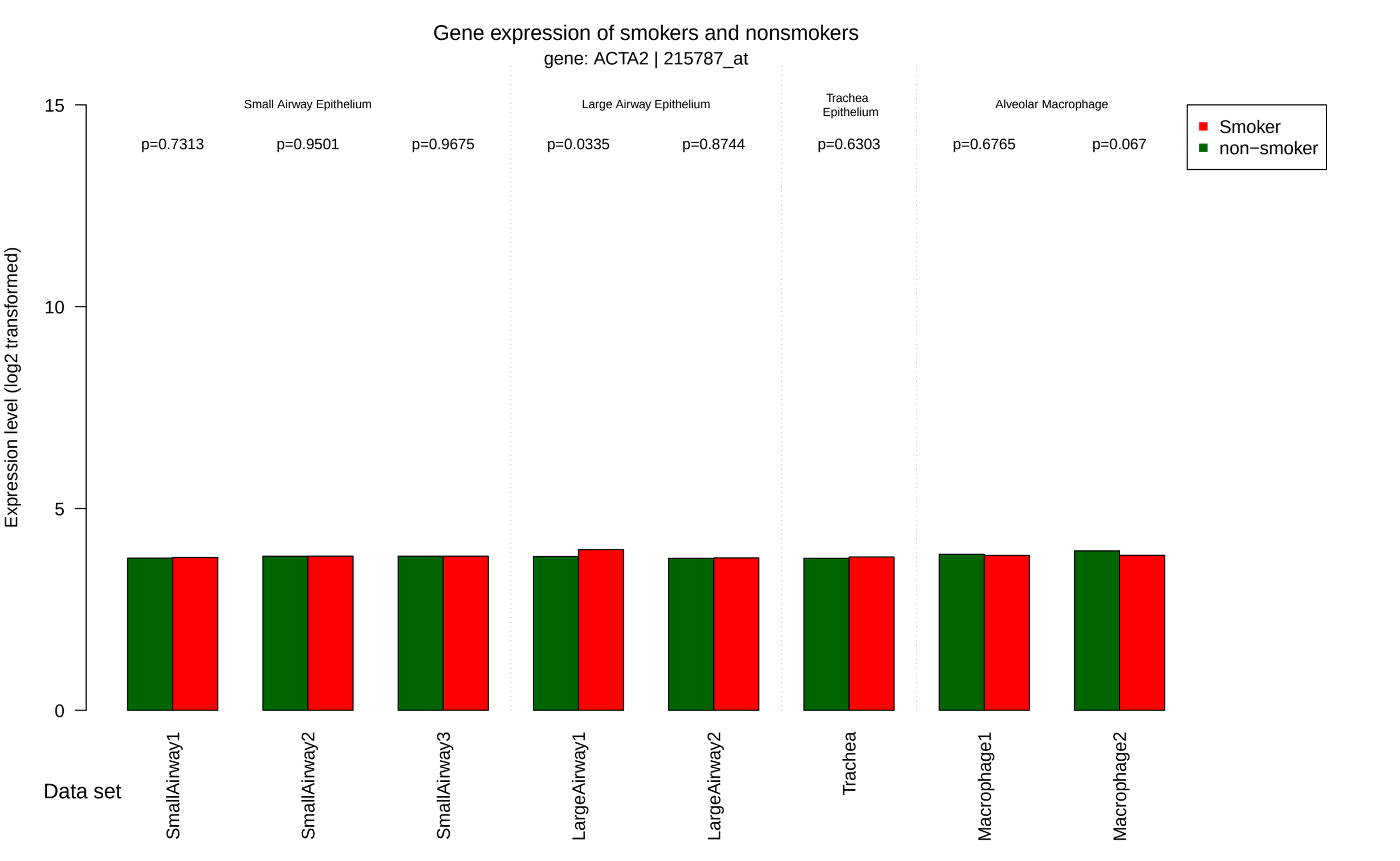
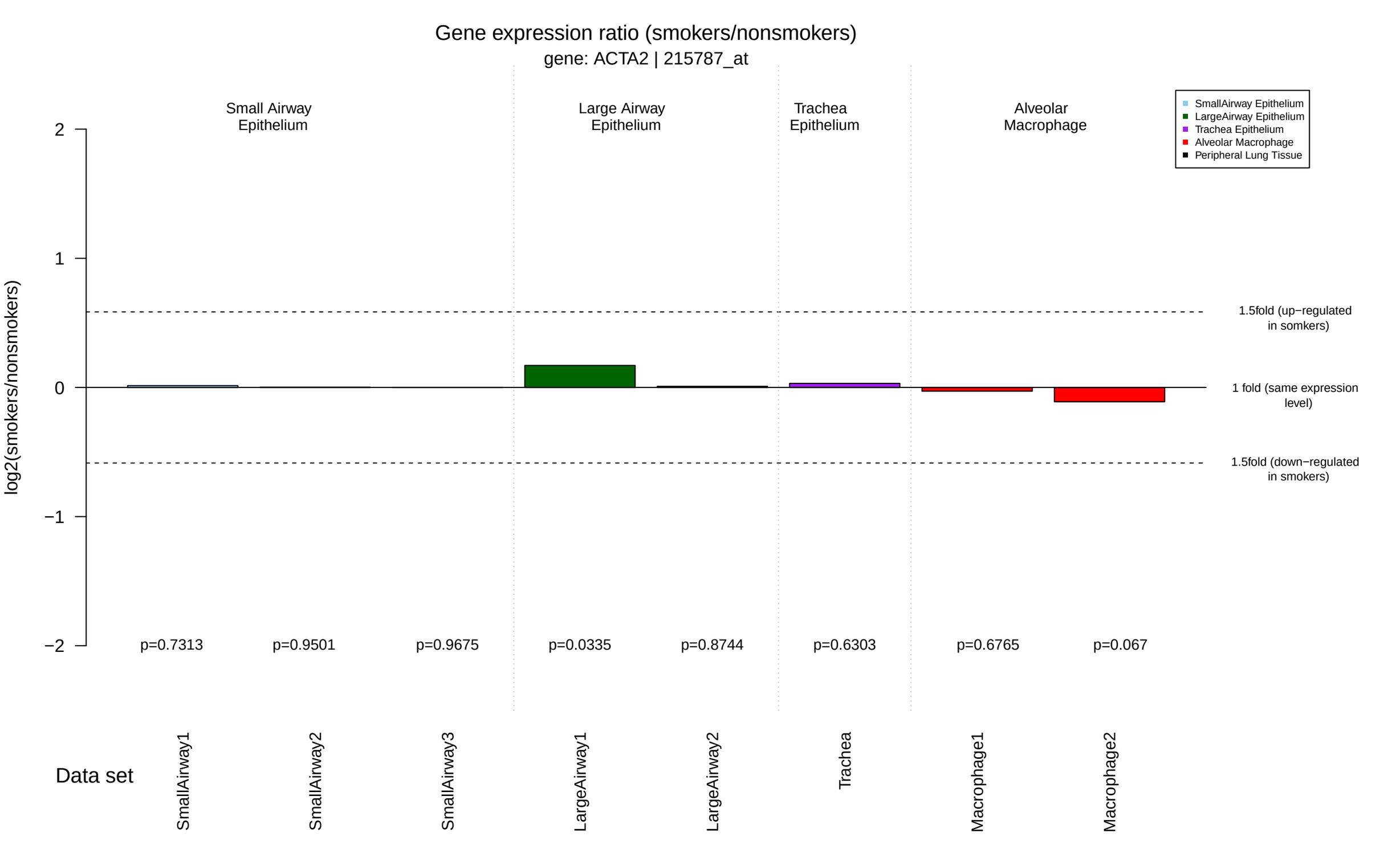
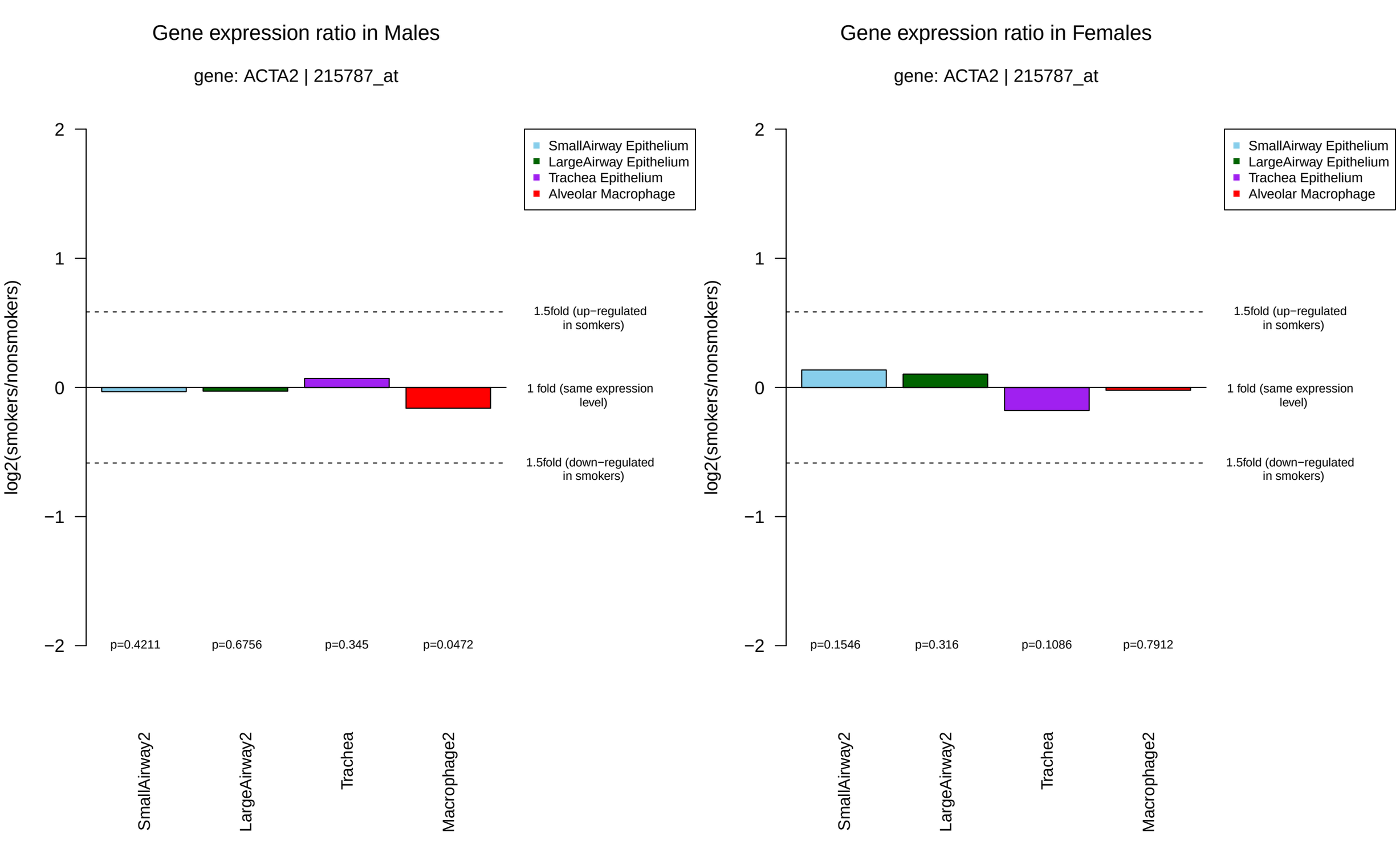
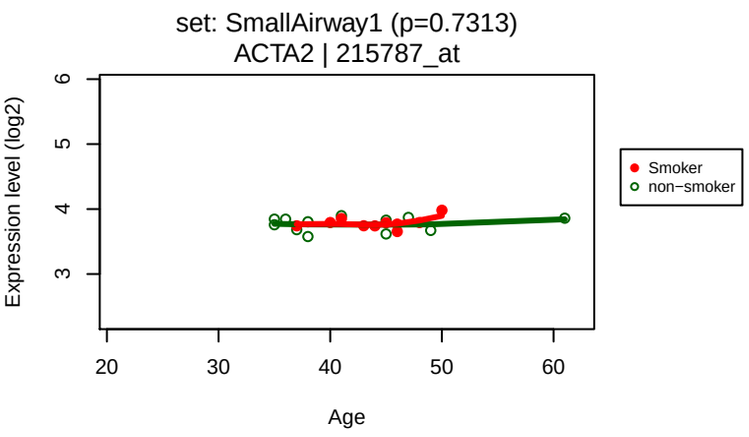
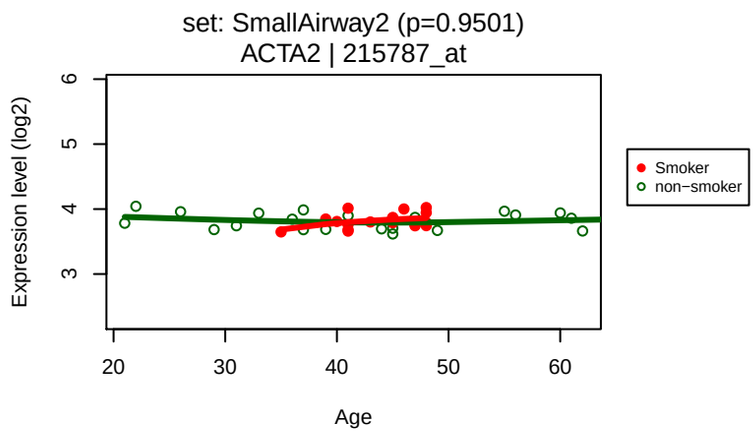
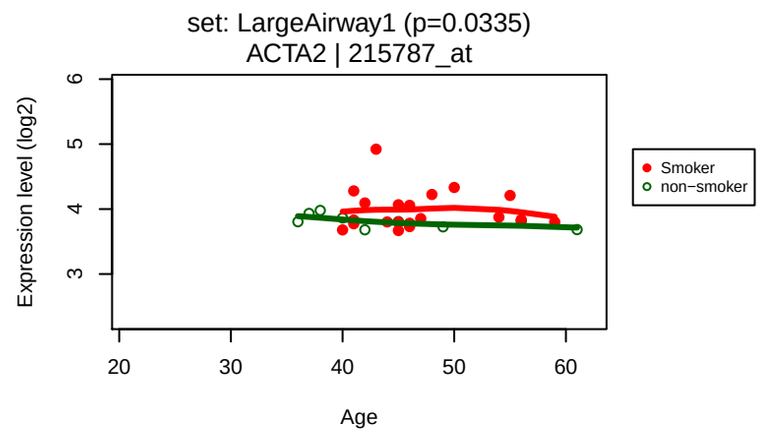

Gene expression of smokers and nonsmokers
gene: ACTA2 | 215787_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.7313
p=0.9501
p=0.9675
p=0.0335
p=0.8744
p=0.6303
p=0.6765
p=0.067
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

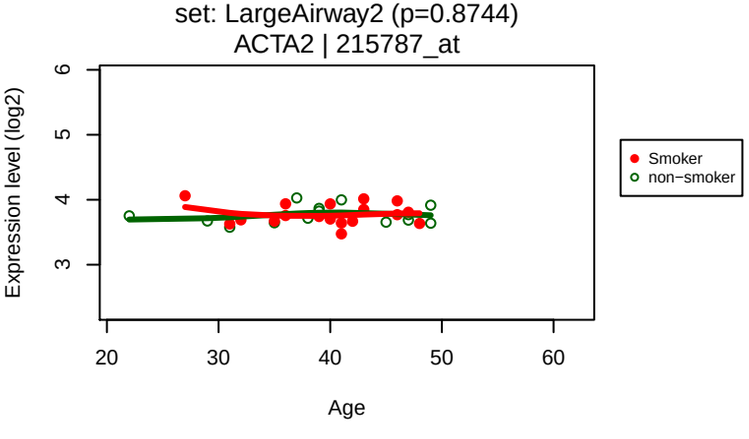

Gene expression ratio (smokers/nonsmokers)
gene: ACTA2 | 215787_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7313
p=0.9501
p=0.9675
p=0.0335
p=0.8744
p=0.6303
p=0.6765
p=0.067
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

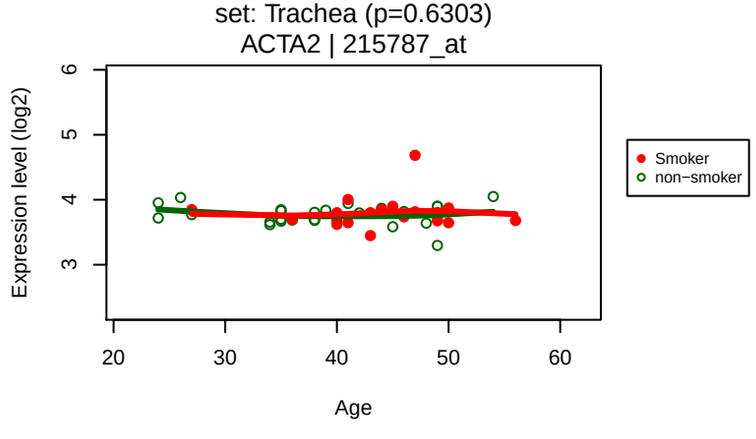

Gene expression ratio in Males
gene: ACTA2 | 215787_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4211
p=0.6756
p=0.345
p=0.0472
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ACTA2 | 215787_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1546
p=0.316
p=0.1086
p=0.7912
SmallAirway2
LargeAirway2
Trachea
Macrophage2

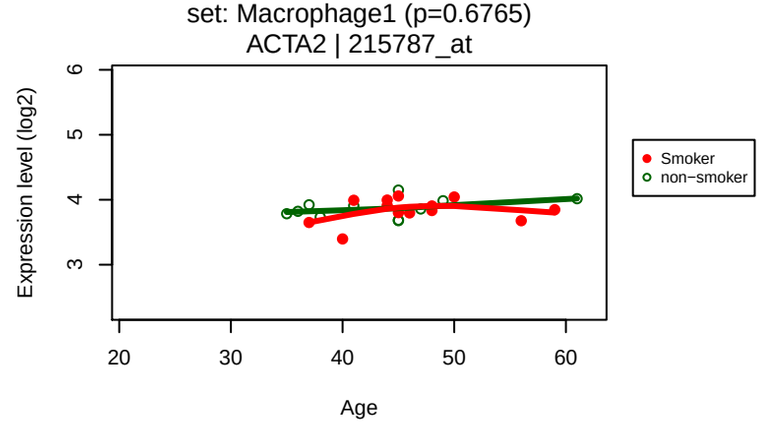

set: SmallAirway1 (p=0.7313)
ACTA2 | 215787_at
Expression level (log2)
Age
Smoker
non−smoker

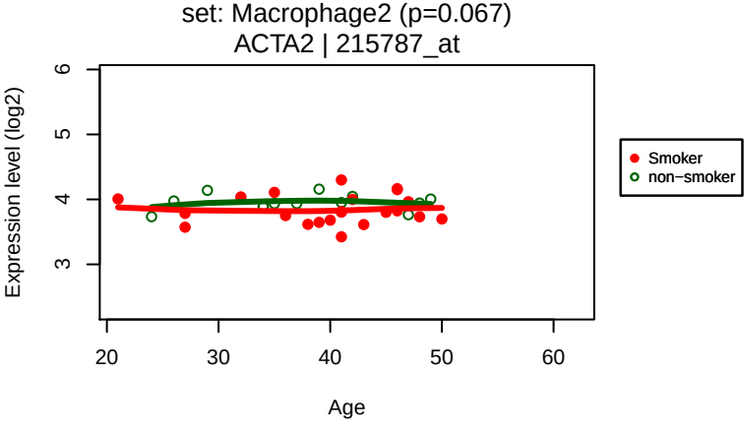

set: SmallAirway2 (p=0.9501)
ACTA2 | 215787_at
Expression level (log2)
Age
Smoker
non−smoker

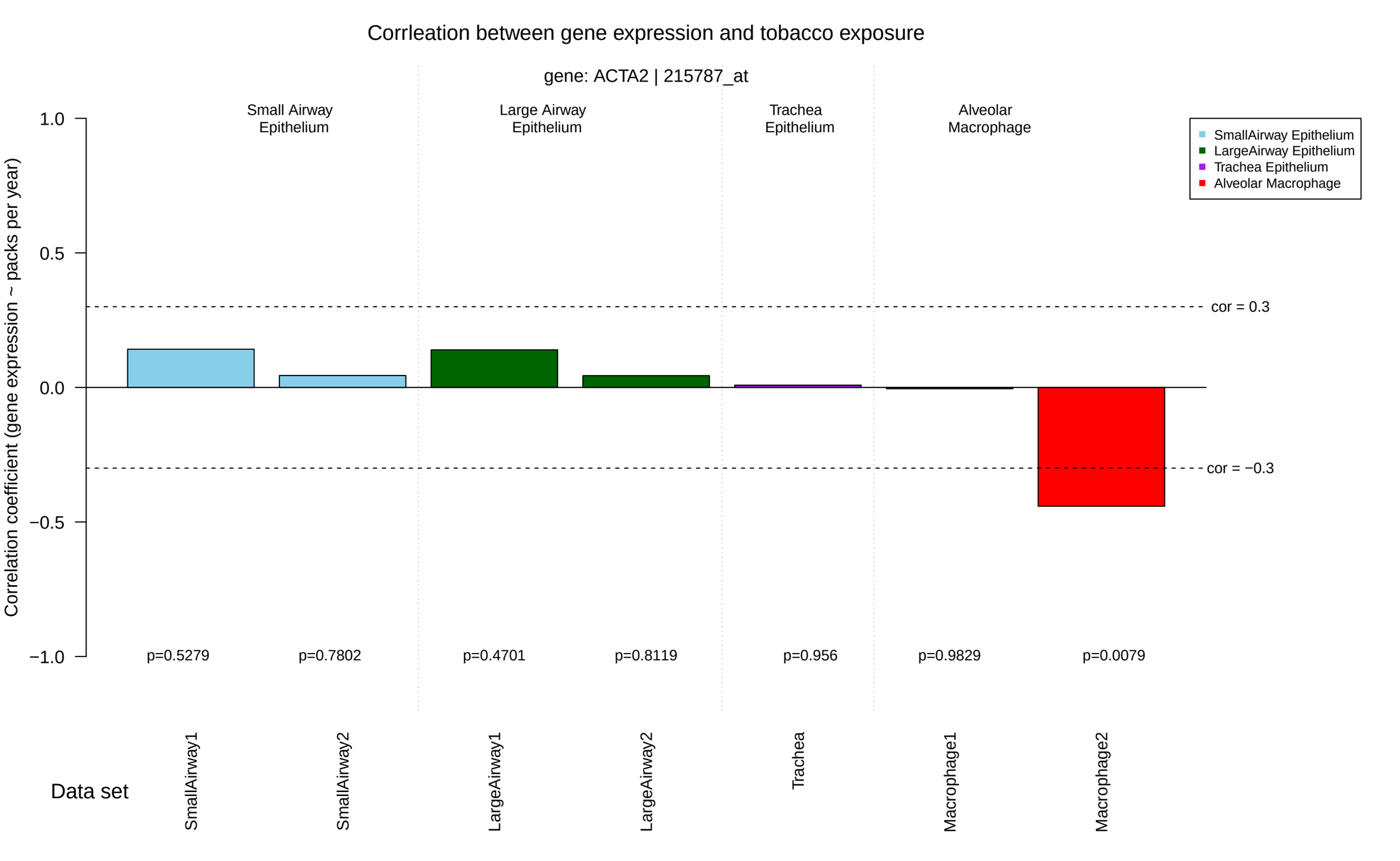

set: LargeAirway1 (p=0.0335)
ACTA2 | 215787_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8744)
ACTA2 | 215787_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6303)
ACTA2 | 215787_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6765)
ACTA2 | 215787_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.067)
ACTA2 | 215787_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ACTA2 | 215787_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5279
p=0.7802
p=0.4701
p=0.8119
p=0.956
p=0.9829
p=0.0079
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2